

A machine-checked proposition set for digital elevation model analysis

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Abstract

This Data Descriptor deposits GeoProofBench v0.1, a versioned catalogue of machine-checkable claims about digital elevation model (DEM) operators. The release contains six concrete first-order propositions, five composition or counter-example records, and forty-four stubbed identifiers reserved for later entries. Each shipped claim has a stable identifier, a one-sentence property, independent Dafny and Lean 4 specifications, a numerical gate, and a recorded verdict. Operators include Wang–Liu pit filling and deterministic eight-neighbour (D8) flow routing. Wolfram exhaustive checks and manim witnesses are deposited where used; they are not required of every identifier. Custom code is supplied only to regenerate those verdicts. The deposit does not introduce a new interpolator. It records where a property holds, where it fails as a property of the operator, and where resampling is expected to break uniqueness. Input rasters include a 5×5 plane, a 256×256 synthetic terrain, labelled synthetic stand-ins, and three public product windows. Data are licensed CC-BY-4.0 and software MIT.

1 Background & Summary

Three decades of DEM software have produced a shared analytical pipeline: pit filling by the Wang–Liu spill rule (interior depressions are raised to the lowest overflow),¹ flow direction, including deterministic eight-neighbour (D8) routing² and flat-surface assignment,³ watershed delineation, and slope or curvature by Horn’s 3×3 weighted fit or Zevenbergen–Thorne profile curvature.^{4,5} Each step is treated as correct in an operational sense that is rarely written so that an independent machine can check it. Independent GIS packages given the same raster can return different flow fields and watershed maps.⁶ Algorithm choice still changes sink handling, extracted networks, and river profiles.^{7–9} That disagreement is not an academic curiosity: catchment area, stream length, and outlet assignment feed flood mapping, infrastructure siting, and published basin statistics. A single undeclared flat ring is enough to invert a basin. GeoProofBench exists because those operators already circulate as data products; what has been missing is a citable record of the obligation each operator is supposed to meet.

The primary audience is the DEM and hydrology community that produces and reuses those operators. Formal methods are the recording language, not the subject of the deposit. Adjacent fields already treat numerical kernels as proof obligations.^{10–17} DEM surfaces have also been used

as input to linear temporal logic path planners for ground vehicles.¹⁸ That line of work verifies a mission on a terrain, not the terrain operators themselves. GeoProofBench is complementary: it archives the operator claims that a later planner, model, or GIS tool may assume.

GeoAI and hydrology question-answering benchmarks score agreement with an annotation or a textbook answer.^{19,20} Those resources remain useful for language-level evaluation. GeoProofBench does not replace them. It supplies the missing layer underneath: a proposition identifier that a model or a GIS package can cite when it claims a geometric guarantee, rather than when it matches a labelled answer. The same identifiers are named obligations that a later automated reasoner, or a GeoMath-style agent that cites DEM operators, can be scored against. This release does not include a trained agent.

GeoProofBench is first a *dataset*: a versioned set of proposition records, rasters, metric tables, and verdicts. Data Records lists the eleven shipped records (six first-order operators and five composition or counter-example records) together with the deposit file layout. Dafny, Lean 4, and Python files regenerate the required gates. Wolfram exhaustive checks and manim witnesses are optional deposits, not part of every record. None of that code is a substitute for the catalogue. Identifiers GPB-006 to GPB-014 and GPB-016 to GPB-050 are stubbed placeholders for community entries; they are not implemented claims. A related remote-sensing article²¹ does not share data with this deposit.

2 Methods

2.1 Proposition model

Each shipped claim is checked on three independent tracks. Dafny and Lean 4 state properties of the abstract operators (fill, D8, slope, curvature) and prove lemmas about those specifications. The lemma counts quoted later are proofs about those specs; they do not certify that a particular Python file is the unique implementation. The reference driver is a canonical Python re-implementation written against the spec. Numerical gates check that this re-implementation, on the deposited arrays, satisfies the spec’s witness. The Python file is not the spec; the spec is the Dafny and Lean file. The gates are finite tests, not proofs. A record is labelled PASS only when the numerical gates hold *and* the corresponding proof files verify. The two provers are independent restatements of the same obligation; neither uses the Python driver as a test oracle for the other. Wolfram exhaustive checks on small grids and manim witnesses are deposited only where used.

Every shipped claim has a stable identifier, a one-sentence property, a Dafny specification, an independent Lean 4 restatement, a Python numerical gate, and a recorded verdict. The identifier, the property sentence, the rasters, the metric JSON, and the verdict constitute the data record. The proof and driver files are the regeneration procedure.

Further labels are not interchangeable with PASS:

- FP (failure of property): the claim is asserted on that input class, but the operator as specified does not satisfy it. The failure is unexpected relative to the hypothesis. It is not a software crash. This release records no FP instance. Version 0.1 ships obligations that hold on their stated class, plus documented counter-examples and one documented boundary

(FAIL-TOLERANCE). An FP would be recorded if a claim that is asserted on a class then failed on an instance of that class.

- **FAIL-TOLERANCE:** the input lies outside the claim’s stated precondition, so the property is not asserted. It is not a licence for the operator to be wrong, and it is not an unexpected failure (that would be FP). In this release it is used only for the four 45° cells of P-COMP-4. Unique D8 after resampling is claimed for a west-sloping plane at 0° and 90° . At 45° , bilinear interpolation mixes cardinal neighbours, so unique D8 is not claimed. That 45° mixing is a known GIS resampling boundary, not a GeoProofBench discovery. Those cells are excluded from the PASS tally and are not relabelled software FAIL.
- **NEGATIVE-RESULT PASS:** a counter-example that is itself the intended record (unfilled four-ring; Zevenbergen–Thorne versus Horn).

2.2 First-order operators

Six operators are complete in this release (Table 5). Pit filling follows the Wang–Liu spill rule:¹ interior pits are raised to the lowest overflow elevation. After that fill, including its spill and index tie-breaking, D8 routing (O’Callaghan and Mark’s deterministic eight-neighbour rule) selects a unique steepest downslope neighbour² under a strict-descent rule: a successor is recorded only when that neighbour is strictly lower than the centre. If no strictly lower neighbour exists, the cell is labelled NoFlow: it has no D8 successor. Watershed uniqueness is stated only after filling, under that strict descent. The 5×5 teaching plane drops elevation by 1 at each flowing step because $A = 1$ and $w = 1$; P-COMP-1b has no descent because the unfilled ring is flat. Zevenbergen–Thorne profile curvature⁵ and Horn second-order slope fitting⁴ are specified separately; they are not treated as interchangeable (see Technical Validation).

2.3 Composition

P-COMP-1 states: after Wang–Liu filling, every interior cell under D8 has exactly one terminus. The argument uses five lemmas: filling yields $\min(\text{nbr}) \leq \text{cell}$; D8 then descends; interior orbits pigeonhole within $n \cdot m$ steps; a rectangular boundary supplies a no-flow terminus; uniqueness collapses candidate termini. P-COMP-1b is the complementary witness: an unfilled four-cell flat ring does not terminate. P-COMP-2 instantiates the same chain on a 256^2 constant plane and on row slopes $\varepsilon \in \{10^{-6}, 10^{-4}\}$. P-COMP-4 resamples a west-sloping plane at scales $\{0.5, 1, 2, 4\} \times$ and rotations $\{0^\circ, 45^\circ, 90^\circ\}$. Its precondition is rotation $\in \{0^\circ, 90^\circ\}$; unique D8 after resampling is claimed only on those angles. The four 45° cells lie outside that precondition: bilinear interpolation mixes cardinal neighbours. That mixing is a known GIS resampling boundary, not a GeoProofBench discovery. P-COMP-5 checks idempotence of filling and hash agreement of one-dimensional fill outputs across Python, Dafny, and Lean. Idempotence $\text{fill}(\text{fill}(h)) = \text{fill}(h)$ is claimed for NoData-free rectangular rasters with finite real numbers.

Fig. 1 is the teaching stencil for this chain. Panel a is the unfilled four-cell flat ring (P-COMP-1b): D8 circulates and does not terminate. Panel b is the 5×5 teaching plane $h(p, q) = Ap + Bq + C$ with $A = 1$ (elevation drop per cell toward decreasing column index), $B = 0$, and $C = 12$ (centre

elevation); (p, q) are integer offsets from the grid centre and the cell width is $w = 1$. After filling, flow is uniquely west-bound. Panel c diagrams the five-lemma order recorded by the Dafny and Lean files; it is a map of the proof, not an extra numerical gate. The multi-resolution stacks and the public windows are Fig. 2 and Fig. 3; they are not this stencil.

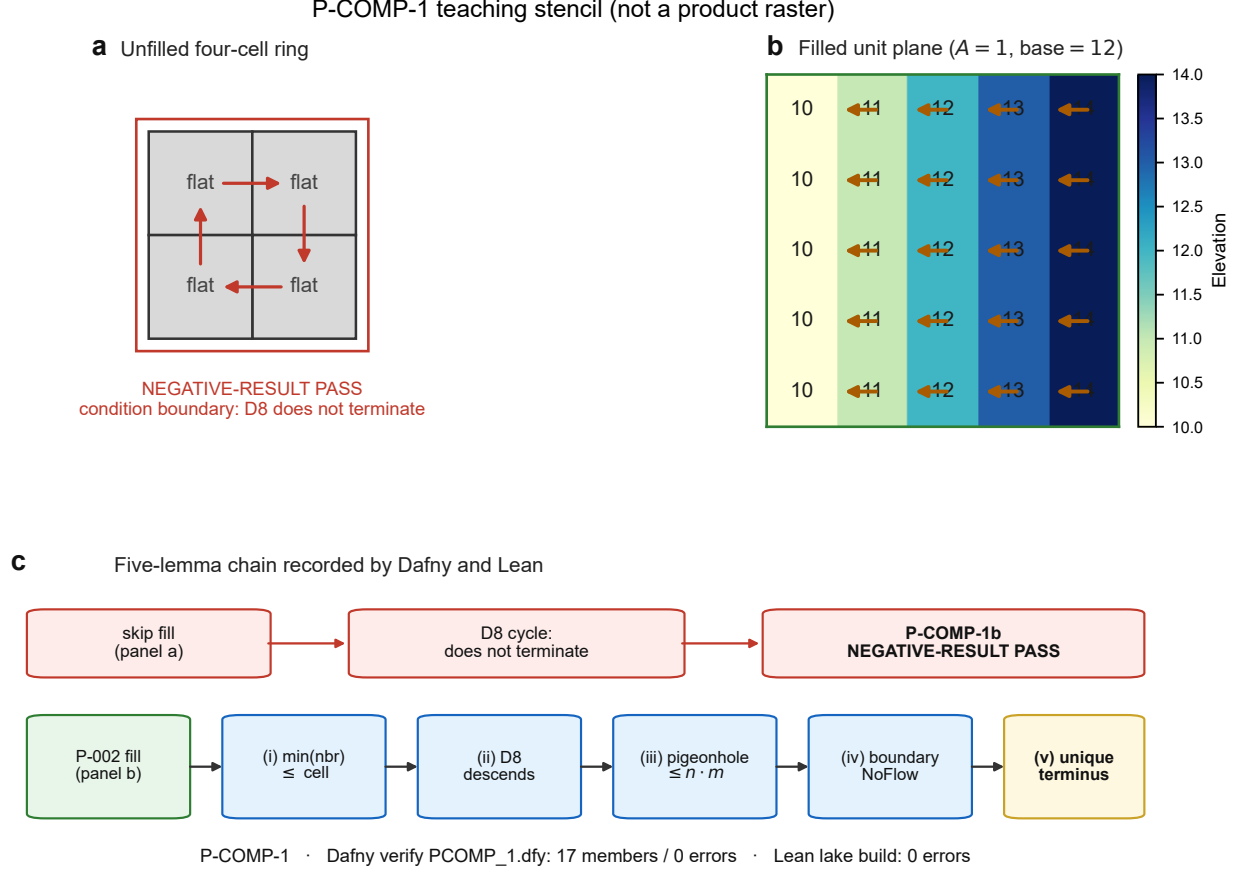


Fig. 1. Teaching stencil for P-COMP-1. (a) Condition-boundary counter-example (P-COMP-1b): unfilled four-cell flat ring; D8 circulates and does not terminate. (b) Filled 5×5 teaching plane $h = Ap + Bq + C$ with $A = 1$, $B = 0$, $C = 12$. Gold arrows mark unique west-bound D8 successors. (c) Five-lemma chain recorded by Dafny and Lean. A cloud `dafny verify` of `PCOMP_1.dfy` reports 17 verified members / 0 errors for that compilation (file plus includes); Lean `lake build` reports 0 errors. Panel a and P-COMP-3 are NEGATIVE-RESULT PASS records of different kinds.

2.4 Input rasters

Four stacks used in the multi-resolution harness are documented in Table 6. Generation parameters for the three synthetic arrays are in Table 1; the driver is `p_comp_1_multires.py` under `experiments/phase2/`. That harness needs four grid sizes, including a 3601^2 SRTM-shaped array and a 4 m coarsened 256^2 array. Intact public files at those two sizes were not obtained from the author workstation. That is a retrieval limit of this release, not a defect of SRTM or 3DEP. Copernicus GLO-30 on tile N32E110 and the 3DEP ImageServer windows (Table 7, Fig. 3) were obtained and are the public-product checks. They do not replace the 3601^2 size-stress raster: a 256^2 GLO-30 window is not a 3601^2 SRTM tile. Version 0.1 therefore deposits labelled synthetic stand-ins for those two sizes. A later version may add a true 3601^2 SRTM tile; until then the

stand-ins are not a product-accuracy claim. The 1 m and 5 m windows were taken from the 3DEP ImageServer nearest-neighbour export of those product families, not from staged COG files (endpoints in Data Availability). The three public products are a USGS 3DEP 1 m lidar-derived window (Griffith Park, Los Angeles),²² a USGS 3DEP 5 m Alaska IFSAR window (Fairbanks area),²³ and a Copernicus DEM GLO-30 window on tile N32E110.²⁴

Noise, when injected in other gates, uses $\sigma \in \{0, 10^{-3}, 10^{-2}, 10^{-1}, 0.5\}$ m.

Table 1. Synthetic raster generators (not product accuracy). All three stand-ins are rebuilt by `p_comp_1_multires.py`. Gaussian σ is in metres of elevation.

Stack	Generator
terrain-A (256 ² , 5 m)	Ramp $0.05x + 0.02y$, two Gaussian hills, twelve carved pits, then $\mathcal{N}(0, 5 \times 10^{-3})$ with NumPy integer seed 20260906 (<code>numpy.random.default_rng</code>).
SRTM-scale (3601 ² , 30 m)	Used after the N32E110 <code>.hgt.gz</code> tile could not be retrieved intact. $1180 + 0.035(3600 - x) + 85 \sin(x/180) + 55 \cos(y/220) + 25 \sin((x + y)/95)$ plus $\mathcal{N}(0, 1.6)$ with integer seed 11032.
lidar-down (256 ² , 4 m)	Fine 1024 ² field $0.04x + 0.015y + 4 \sin(x/18) \cos(y/22) + 1.2 \sin(x/3.5) + 0.8 \cos(y/4.2)$ plus $\mathcal{N}(0, 0.25)$ with integer seed 20260907, then 4×4 block mean to 256 ² .

2.5 Verification environment

Numerical gates run on Python 3.12 (NumPy) on the author workstation. Dafny 4.11 and Lean 4.18.0 with mathlib are re-executed from the deposited sources on an independent AutoDL cloud mirror that does not share the workstation’s verifier cache; the lemma counts quoted below are those cloud logs, not copies of local output. Wolfram Script 14 is used only on the workstation (the cloud mirror has no Wolfram licence; that skip is recorded, not treated as FAIL). Commands and file paths live in the deposit under `experiments/` and `formal/`.

2.6 Ethics

No human participants, animals, or personal data were involved. Input rasters are synthetic arrays or publicly distributed elevation products.

3 Data Records

While the formal specifications provide the highest recorded assurance, the core data product—proposition statements, input rasters, metric tables, and verdicts—can be used and cited independently of Dafny or Lean. The citable object is the proposition catalogue `GeoProofBench-v0.1-deposit.zip` at Science Data Bank.²⁵ Table 3 lists the eleven shipped records. Forty-four stubbed identifiers (GPB-006 to 014 and GPB-016 to 050) are reserved names in `benchmark/`; they are planned extensions, not implemented claims in v0.1, and they are not rows of that table. Table 4 is the deposit file layout. Data Records describes the JSON, rasters, and metric tables; Code Availability lists the Dafny, Lean, and Python regeneration files.

Metrics JSON live under `experiments/phase2/results/` as `gpb021` (P-COMP-1), `gpb024*` (multi-resolution), `gpb025*` (resampling), `gpb026*` (hashes), and `gpb027_realworld`. Table 2 is

the field dictionary. In Table 6 the column n_{out} is the JSON field `uniq_out`. Table 7 lists the three public 256^2 product windows. Unless a suite states otherwise, `n_term` and `uniq_out` are computed on the interior frame (excluding a one-cell boundary).

Table 2. Dictionary for metrics JSON fields used in this Descriptor.

Field	Type	Unit	Definition
<code>n_pit</code>	integer	—	Count of interior 4-neighbour local minima on the unfilled raster.
<code>n_term</code>	integer	—	Count of interior cells whose D8 orbit reaches a terminus.
<code>uniq_out</code>	integer	—	Number of distinct termini among those terminating cells.
<code>longest</code>	integer	cell steps	Length of the longest terminating D8 path.
<code>ridgeline_frac</code>	float	1	Fraction of interior cells with D8 in-degree 0.
<code>drainage_density_per_m</code>	float	m^{-1}	Count of interior cells with in-degree ≥ 2 , divided by (interior cell count $\times$ cell size).

Table 3. Shipped GeoProofBench v0.1 records. Companions are noted, not counted as extra first-order or composition identifiers.

Status	Count	Identifiers	Note
First-order	6	GPB-001 to 005 and GPB-015	GPB-002b is the 2-D companion of 002 (Table 5); it is not a seventh first-order identifier
Composition	5	P-COMP-1 to 5	P-COMP-1b is the unfilled-ring companion of 1

Table 4. Deposit file layout (data products and regeneration sources). SHA-256 hashes of the metrics JSON are in the named result files; Lake caches are omitted.

Path	Contents
<code>experiments/</code>	Metrics JSON: <code>gpb021</code> , <code>gpb024*</code> , <code>gpb025*</code> , <code>gpb026*</code> ,
<code>phase2/results/</code>	<code>gpb027_realworld</code> .
<code>experiments/*/figures/</code>	Optional mp4 witnesses where deposited.
<code>benchmark/</code>	Identifier list, including forty-four planned-extension stubs.
<code>papers/P2-geoproofbench/</code>	Descriptor text and figures.
<code>formal/dafny/*.dfy</code>	13 Dafny specification files (Code Availability).
<code>formal/lean4/</code>	VeriGIS library, <code>lakefile.toml</code> , <code>lean-toolchain</code> (Code Availability).

Table 5. Shipped first-order operators and lead artefacts. GPB-002b is the 2-D companion of GPB-002 and is not a seventh first-order identifier. The six first-order records are GPB-001 to 005 and GPB-015. GPB-015 uses the historical file prefix P006 (watershed module); the identifier is 015, not a mismatch with the six-record count.

ID	Property	Lead Dafny file
GPB-001	Slope sign on a plane	P001_horn_slope.dfy
GPB-002	Pit filling to spill	P002_pit_filling.dfy
GPB-002b	2-D fill monotone, idempotent	P002_pit_filling_2d.dfy
GPB-003	ZT curvature zero on a plane	P003_curvature.dfy
GPB-004	Horn fit exact on a plane	P004_consistency.dfy
GPB-005	Unique D8 successor	P005_d8.dfy
GPB-015	Watershed uniqueness under D8	P006_watershed.dfy

Table 6. Multi-resolution stacks used with the same fill-then-D8 chain. Asterisks mark synthetic stand-ins, not public elevation products.

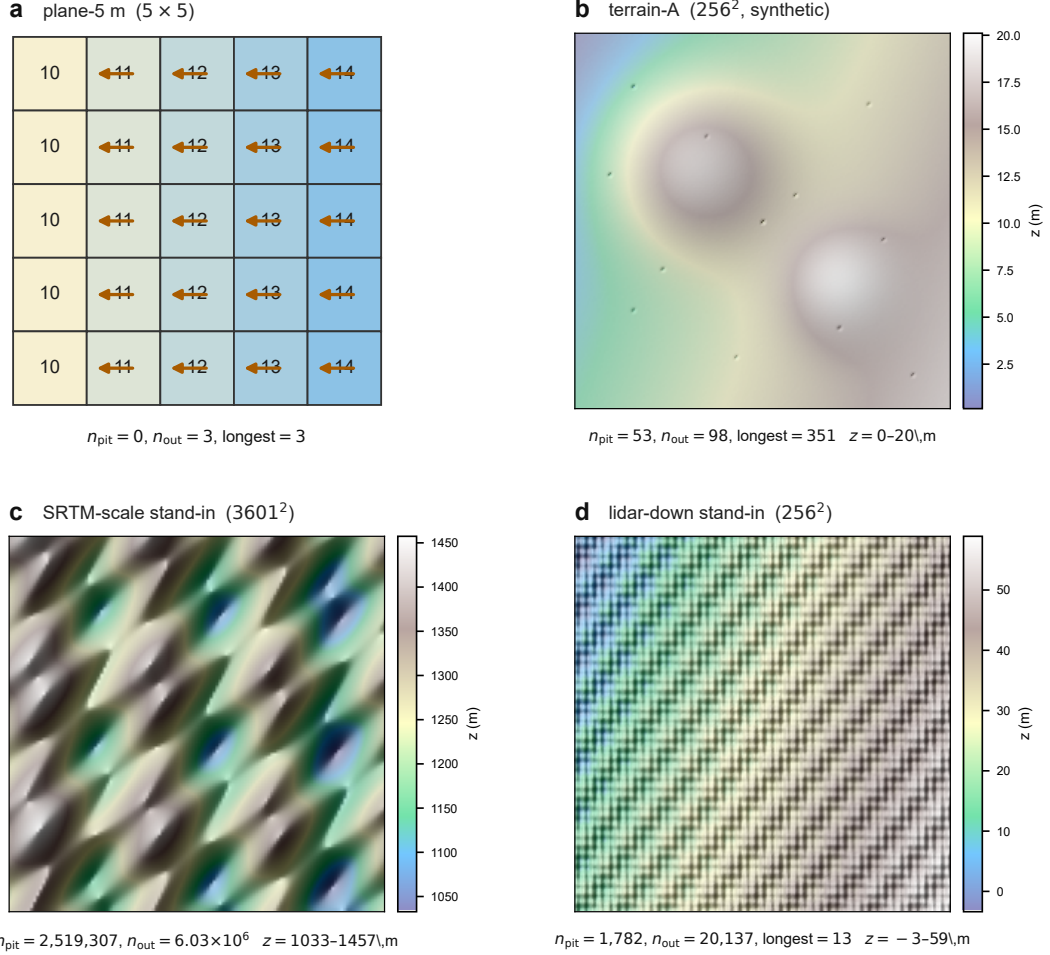
Stack	Grid	n_{pit}	n_{term}	n_{out}	longest
plane-5 m	5×5	0	9	3	3
terrain-A	256^2	53	64516	98	351
SRTM-scale stand-in*	3601^2	2519307	12952801	6027216	28
lidar-down stand-in*	256^2	1782	64516	20137	13

Table 7. Public product windows deposited at 256^2 . Columns match the metric JSON; P-COMP-1 gates all PASS. Ridge: interior D8 in-degree 0. D_d : drainage density (m^{-1}).

Window	Cell size	n_{pit}	n_{out}	ridge	D_d
USGS 3DEP lidar, Griffith Park	1 m	88	234	0.125	0.106
Alaska IFSAR, Fairbanks window	5 m	270	253	0.225	0.0363
Copernicus GLO-30, N32E110	28.4 m	626	4981	0.318	0.00685

4 Data Overview

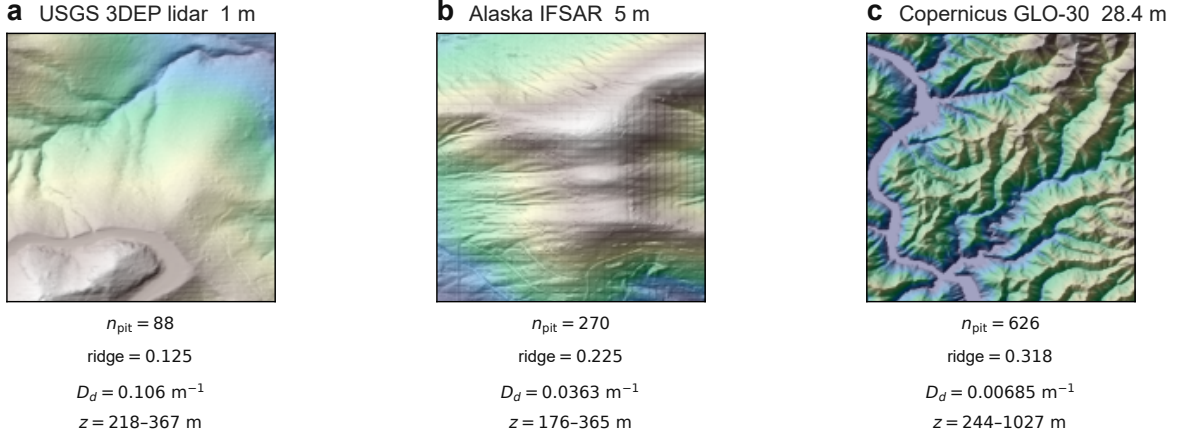
Table 6 and Fig. 2 report fill-then-D8 on four grid sizes. After filling, every interior cell terminates (n_{term} equals the interior count). Outlet count grows with roughness and grid size (3 on the 5×5 plane; 98 on terrain-A; 6.03×10^6 on the noisy SRTM-scale stand-in), while longest paths do not (351 on terrain-A versus 13 on the lidar-down stand-in after 4×4 block mean). Table 7 and Fig. 3 apply the same gates to three public 256^2 product windows; all three PASS P-COMP-1 (zero interior pits after fill; 64 516 / 64 516 interior cells terminate). Ridge fraction rises and drainage density falls as cell size increases from 1 m lidar to 28.4 m GLO-30; that pattern is a characteristic of these windows, not a general landscape claim. Hydrologic users who need a named obligation for unique outlets after filling can start from P-COMP-1; developers who need a named boundary for rotated resampling should use the FAIL-TOLERANCE cells in Usage Notes. Further summary statistics should be computed from the deposited arrays.



Elevation colour is per panel (blue/green: lower; yellow/white: higher). Panel c is a display-only block-mean of the deposited 3601^2 raster. c, d are synthetic stand-ins.

Fig. 2. Fill-then-D8 inventory on the four deposited stacks. **(a)** Unit plane, 5×5 , shown as a labelled grid with D8 arrows (hillshade is undefined on a plane); cell numbers are elevation. **(b)** Synthetic terrain-A, 256^2 . **(c)** SRTM-scale stand-in. The deposited file is the full 3601^2 array; the panel is a display-only block-mean downsample, not a substitute for that raster. **(d)** Lidar-down stand-in, 256^2 ; residual 4×4 block structure after block mean at 4 m is a property of the deposited stack, not a display artefact. In **b–d**, colour maps elevation independently in each panel (blue/green: lower; yellow/white: higher). n_{pit} , n_{out} and longest path are the deposited metrics. Panels **c** and **d** are not public elevation products. Public product windows are Fig. 3.

Public 256² windows: P-COMP-1 gates PASS



Elevation colour is per panel (blue/green: lower; yellow/white: higher).
ridge = interior D8 in-degree 0; $D_d = (\text{interior in-degree} \geq 2) / (\text{interior cells} \times \text{cell size})$.

Fig. 3. Public 256² windows used for diversity checks. (a) USGS 3DEP 1 m lidar, Griffith Park. (b) USGS 3DEP 5 m IFSAR, Fairbanks area. (c) Copernicus GLO-30, tile N32E110. Colour maps elevation independently in each panel (blue/green: lower; yellow/white: higher) and is not a between-product height comparison. Each panel reports interior pit count n_{pit} , ridgeline fraction (interior cells with D8 in-degree 0) and drainage density D_d (interior cells with in-degree ≥ 2 , divided by interior cell count times cell size). All three windows PASS P-COMP-1.

5 Technical Validation

Validation supports the technical quality of the deposited records. It is not an analysis of a new terrain algorithm. Table 8 collects the gates that a user can regenerate from the archive.

Table 8. Summary of recorded verdicts. FAIL-TOLERANCE cells are excluded from the PASS tally. This release records no FP instance.

Record	Verdict
P-COMP-1, 5×5 plane	PASS
P-COMP-1b four-ring	NEGATIVE-RESULT PASS
P-COMP-2	PASS
P-COMP-3	NEGATIVE-RESULT PASS
P-COMP-4	8 PASS + 4 FAIL-TOLERANCE (all 45° rotations)
P-COMP-5 hashes / 2-D fill	12/12 and 4/4 PASS
Three public windows	PASS
FP (this release)	none recorded

5.1 Numerical and formal gates

On the 5×5 unit plane used for P-COMP-1 (Fig. 1b), the record reports: no interior pit after fill; each flowing D8 step drops elevation by 1; all 9 interior cells terminate in at most 3 steps

with 3 distinct outlets. On the author workstation, Wolfram enumeration of 4^4 maps gives `pigeonholeAll=true`, `descentAllFix=true`, `ringHasFixedPoint=false`. The AutoDL cloud mirror has no Wolfram licence and skips that script; Python, Dafny, and Lean gates do not depend on it.

An independent AutoDL cloud re-run of `dafny verify` on the deposited files (no workstation cache) reports 17 verified members / 0 errors on `PCOMP_1.dfy` and 12 / 0 on the termination file `P006_terminate_under_strict.dfy`. The same mirror reports 25 / 0 (`PCOMP_2.dfy`), 20 / 0 (`PCOMP_4_homotopy.dfy`), and 15 / 0 (`PCOMP_5_idempotent.dfy`). Those integers are verifier-reported members for each compilation unit (the named file plus its `includes`). They are not lemma counts of GeoProofBench as a whole. `PCOMP_1.dfy` itself states 11 lemmas: five correspond to steps (i)–(v) of Fig. 1c (`NoPitImpliesDescent`, `D8PreservesDescent`, `OrbitLengthBound`, `BoundaryNonEmpty`, `FillThenWatershed`); the others are 2-D or pigeonhole variants, uniqueness wrappers, and an example. Lean `lake build` of the deposited `VeriGIS` library against the locked `mathlib` was likewise executed on that mirror and completes with 0 errors: the library builds. The lockfile imports 2764 `mathlib` modules; that number is a toolchain size, not a count of GeoProofBench lemmas, and is not used as evidence that the propositions hold.

P-COMP-2: a 256^2 constant plane has 65536 / 65536 terminating cells (all `NoFlow`). Row slopes $\varepsilon = 10^{-6}$ and 10^{-4} terminate with longest path 255 and 256 outlets (all north). The P-COMP-2 metrics JSON reports `visited_cells = 16908288`. That field is a path-length sum, not a unique-cell census, and it is not a column of Table 6. For each start cell the driver adds the D8 path length in steps plus one (the start itself), then sums over all 256^2 starts in each suite. The constant plane contributes 65 536. Each row-slope suite contributes 8 421 376 ($256 \times (255 \times 256/2) + 65\,536$). The three-suite total is 16 908 288.

P-COMP-4: 8 of 12 scale-rotation cells PASS with unique D8 and $\sigma = 0$. All four 45° cells are FAIL-TOLERANCE as defined above (precondition rotation $\in \{0^\circ, 90^\circ\}$). P-COMP-5: four one-dimensional fill instances (plane, pit, slope, cascade) are hashed with SHA-256 in Python, Dafny, and Lean. Agreement on all 4 instances $\times$ 3 implementations is recorded as 12 / 12. Two-dimensional fill(`fill(h)`) = `fill(h)` holds at $\sigma = 0$ on four `NoData`-free rectangular rasters with finite real numbers (4 / 4). A reversed heap-tie schedule is included: Wang–Liu priority flood keys the heap by fill elevation then (r, c) ; reversing the sign of the (r, c) components yields a second schedule. The two outputs agree within the recorded epsilon.

P-COMP-1 on the three public windows: interior pits after fill are 0; 64516 / 64516 interior cells terminate (Table 7).

5.2 Counter-examples retained as records

An unfilled 4-ring is a non-terminating D8 orbit (P-COMP-1b; Fig. 1a): a condition-boundary counter-example that marks where P-COMP-1 does not apply. Two 9×9 patches $z = Dx^2 + Ax$ show that Zevenbergen–Thorne profile curvature $H_{xx} = 2D$ and Horn’s second-order coefficient $D_x = A$ are not the same function (P-COMP-3; Fig. 4). The pairs $(A, D) = (0.21, 0.00705)$ and $(0.07, 0.03485)$ were chosen so that both centre-cell values are nonzero and need not vanish together; they are not random draws. The figure is an existence witness, not a population claim.

Both records are catalogued as NEGATIVE-RESULT PASS.

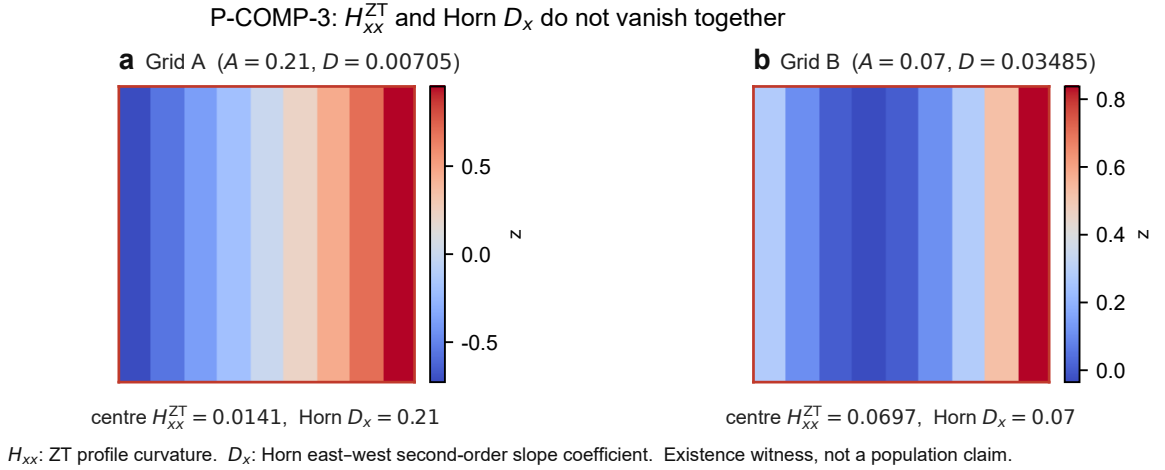


Fig. 4. Operator-inequivalence counter-example (P-COMP-3), a NEGATIVE-RESULT PASS of a different kind from the condition-boundary ring in Fig. 1a. **(a)** Grid A: $z = Dx^2 + Ax$ with $A = 0.21$, $D = 0.00705$. **(b)** Grid B: $A = 0.07$, $D = 0.03485$. H_{xx} is Zevenbergen–Thorne profile curvature; D_x is Horn’s east–west second-order slope coefficient. Centre-cell values are both nonzero and do not vanish together, so the two operators are not interchangeable. The grids demonstrate existence, not a population claim.

6 Usage Notes

6.1 Intended reuse

Three entry points are intended.

DEM algorithm developers can attach a GPB identifier to a slope, fill, D8, or watershed implementation and rerun the deposited Python gate on their output raster. That path is sample testing: it checks finite arrays, not a proof of the new code. A mismatch is a disagreement with a named obligation, not with an unnamed laboratory convention. The documented counter-examples (P-COMP-1b, P-COMP-3) are stress tests for new algorithms: they mark edge cases that a conventional accuracy score on a filled, well-sloped raster can miss.

Hydrologic modellers can cite P-COMP-1 when a workflow claims unique outlets after filling, and can cite P-COMP-1b or the 45° FAIL-TOLERANCE cells when a study area contains flats or rotated resampling. The public-window metrics in Table 7 are starting examples, not regional products.

Formal-methods researchers who need a machine-checked guarantee must restate their operator against the deposited Dafny or Lean specifications and rerun `dafny verify` or `lake build`. The Lean restatement is independent of the Dafny kernel; agreement is a cross-check, not a duplicated script. That embedding is the specialist path; it is not required of GIS users who only rerun the Python gates. Stubbed identifiers GPB-006 to GPB-014 and GPB-016 to GPB-050 are the extension surface.

6.2 How to run

From the deposit root, each operator has a `run_*.sh` (or the matching `.py` on Windows). Formal files verify with `dafny verify formal/dafny/<file>.dfy` and `cd formal/lean4 && lake build`. Lean 4.18.0 is pinned by `formal/lean4/lean-toolchain`. Lake caches are not deposited; the first `lake build` fetches mathlib from the lockfile (network required; a cold cache takes on the order of tens of minutes). Python drivers import NumPy and, for the public windows, GDAL; those imports are the deposited dependency list. Wolfram scripts are optional. Do not treat a missing Wolfram licence as a failed gate.

The 45° resampling cells and the synthetic SRTM / lidar-down rows are intentional boundaries. GPU parallel flow accumulation is not in this release.

7 Data Availability

The Data Records section lists files and formats. The version of record is openly available at Science Data Bank, V1:²⁵ <https://doi.org/10.57760/sciencedb.011r9> (CSTR <https://cstr.cn/31253.11.sciencedb.011r9>; identifier 31253.11.sciencedb.011r9). Download is unrestricted. Data are licensed CC-BY-4.0 and software MIT.

Public rasters used as input were fetched as follows. USGS 3DEP windows come from the ImageServer export <https://elevation.nationalmap.gov/arcgis/rest/services/3DEPElevation/ImageServer/exportImage> with `bboxSR=4326, imageSR=4326, size=256,256, format=tiff, interpolation=RSP_NearestNeighbor`: Griffith Park lidar (`west=-118.2964, south=34.1348, cell 1 m`)²² and Fairbanks IFSAR (`west=-147.815, south=64.894, cell 5 m`).²³ Copernicus GLO-30 tile N32E110²⁴ was read with GDAL `/vsicurl/` from the public COG https://copernicus-dem-30m.s3.eu-central-1.amazonaws.com/Copernicus_DSM_COG_10_N32_00_E110_00_DEM/Copernicus_DSM_COG_10_N32_00_E110_00_DEM.tif.

The work uses synthetic rasters and public elevation products only. It does not involve human participants, personal data, or restricted geographic holdings.

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9 Code Availability

Custom code is included in the same archive as the data (MIT licence). Dafny specifications live under `formal/dafny/` (13 files); Lean 4 restatements under `formal/lean4/` (`lakefile.toml`, `lean-toolchain`; Lake caches omitted); Python and optional Wolfram drivers under `experiments/`. Software versions used to generate the present verdicts: Python 3.12, Dafny 4.11, Lean 4.18.0 with mathlib, NumPy, GDAL for windowed GeoTIFF reads, manim 0.21 for animations. Public rasters are those listed under Data Availability. The same tree is at <https://github.com/fariell/verifiable-geocomputation> (public development copy; Scientific Data is not double-blind). Reviewers

can download code from the unrestricted Science Data Bank zip at the Data Availability DOI; a separate anonymised GitHub mirror is not required. No additional proprietary binaries are required for the numerical gates.

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